F78PA Probability & Statistics A

Tutorial 5

1. Suppose that X is a normal random variable with parameters $\mu = 4$ and $\sigma^2 = 25$. Use statistical tables to find (a) $\mathbb{P}(X > 6)$ (b) $\mathbb{P}(X < 2)$ (c) $\mathbb{P}(0 < X < 10)$
2. Suppose that $X \sim \mathrm{N}\left(\mu, \sigma^2\right)$ is a normal random variable with $\mu = 5$. If $\mathbb{P}(X \geq 9) = 0.3$, determine the value of $\sigma^2 = \mathbb{V}\mathrm{ar}[X]$.
3. Suppose that the time, T, that it takes a student to travel from home to university is normally distributed with mean 40 minutes and standard deviation 7 minutes. At what time should the student leave home in order to ensure that the probability that they are on time for a lecture that starts at 9.20 am is at least 0.95?
4. Suppose that the random variable X is uniformly distributed on the interval $[0, 1]$, that is, $X \sim \mathrm{U}(0, 1)$. For each of the random variables defined below, determine its image and find its distribution function. In each case, determine whether the corresponding density function exists (giving reasons for your answer). If it exists, determine the density.

   (a) $Y = \dfrac{1}{\sqrt{X+3}}$

   (b) $W = \max\{1, -\log X\}$

   (c) $V = -\log(X)$

5. (a) Give an example of a distribution function that is not continuous. Explain why you can be certain that your example is a distribution function.

   (b) Give an example of a function that takes values between 0 and 1 but which is not a distribution function. Explain why you can be certain that your example is not a distribution function.

6. A bus travels between two cities, A and B, which are 120 miles apart. There is a bus service station in city A, in city B, and a further service station located halfway between cities A and B. If the bus has a breakdown, the distance in miles from the breakdown to city A has a uniform distribution over the interval $[0, 120]$, and the bus then travels to the nearest service station. In the event of a breakdown, find the expected distance to the nearest service station.
7. Suppose X is a random variable with continuous distribution function F_X and density function f_X. Also, suppose that $Y = aX + b$, where $a, b \in \mathbb{R}$ and $a > 0$.

   (a) Express the distribution function F_Y of Y in terms of the distribution function F_X, and express the density function f_Y of Y in terms of f_X.

   (b) How does your answer in part (a) change if $a < 0$?

   (c) An archer shoots an arrow at a circular target of radius 1 metre in such a way that the probability that the the arrow hits a region A on the target is proportional to the area of the region A. We also suppose that the archer never misses the target. Let X denote the distance, in centimetres, between where the arrow hits the target and the centre of the target, and suppose now that the archer's score is computed as $Y = 5 - \frac{X}{20}$.

   Determine $\mathrm{Im}(Y)$, the image of Y, and use part (b) above to determine f_Y, the density of Y.

8. Suppose that $Z \sim \mathrm{N}(0, 1)$. Show that $\mathbb{E}[Z] = 0$ and $\mathbb{V}\mathrm{ar}(Z) = 1$.
9. Suppose that $Z \sim \mathrm{N}(0, 1)$ and let $Y = Z^2$. Show that the random variable Y has a $\Gamma(\frac{1}{2}, \frac{1}{2})$ distribution. (Note: this is a special distribution which is also known as a χ^2 distribution with one degree of freedom, $Y \sim \chi^2(1)$.)

Hint: $\Gamma(1/2) = \sqrt{\pi}$ and you should show that Y has a pdf given by

$$f_Y(y) = \begin{cases} \frac{2^{-1/2}}{\Gamma(\frac{1}{2})} y^{-\frac{1}{2}} e^{-y/2} & y \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

Exam Style Question

Consider the function

$$f_X(x) = \begin{cases} x, & \text{if } 0 \leq x \leq 1, \\ \frac{1}{2}, & \text{if } 1 < x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Sketch the graph of the function f_X and explain why it is a density function for a random variable X. [2 marks]

(b) State the image, $\text{Im}(X)$, of X. [1 mark]

(c) Find the expected value, $\mathbb{E}[X]$, of X. [2 marks]

(d) Find the cumulative distribution function, F_X, of X. [3 marks]

(e) Suppose that $Y = e^X$.

i) Explain why $\text{Im}(Y) = [1, e^2]$ and show that for $1 \leq y \leq e^2$

$$F_Y(y) = F_X(\log(y))$$

where F_Y is the cumulative distribution function of Y. [3 marks]

ii) Use part ei) above to show that the density for Y is given by

$$f_Y(y) = \begin{cases} \frac{\log(y)}{y}, & \text{if } 1 < y \leq e, \\ \frac{1}{2y}, & \text{if } e < y \leq e^2, \\ 0, & \text{otherwise.} \end{cases}$$

[3 marks]

[Total 14 marks]